Pharmaceutical prescribing intelligence

Independent, AI-assisted portfolio project

- England · atorvastatin · 36 observed months
- SQL, statistical diagnostics, frozen forecasts and decision support
- Public administrative evidence; no proprietary client data

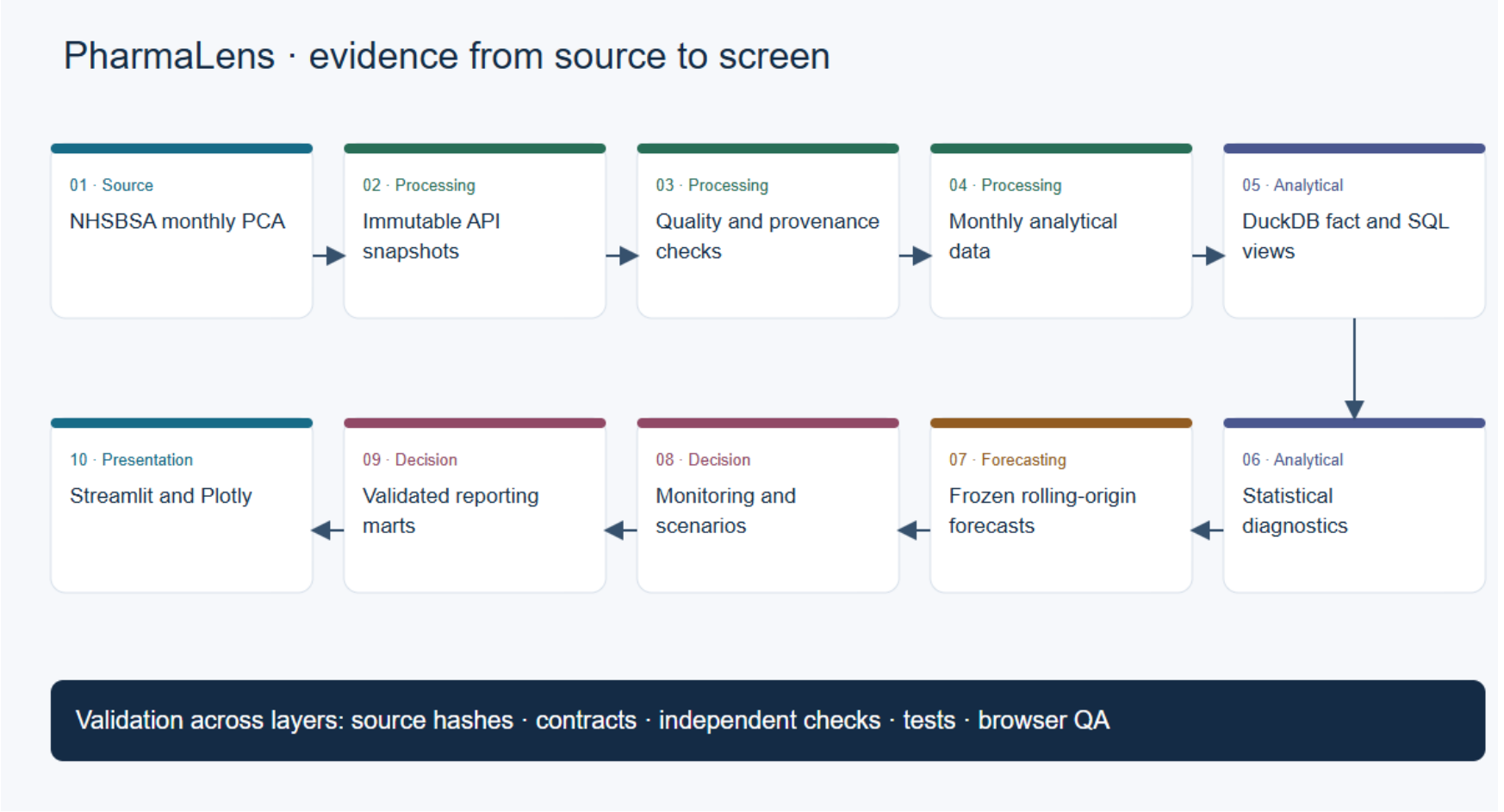
Make the evidence useful without overstating it

- What changed in observed activity and ingredient cost?
- Which forecasts and review priorities need caution?
- What follows from an explicit accounting assumption?

Preserve grain, units and provenance

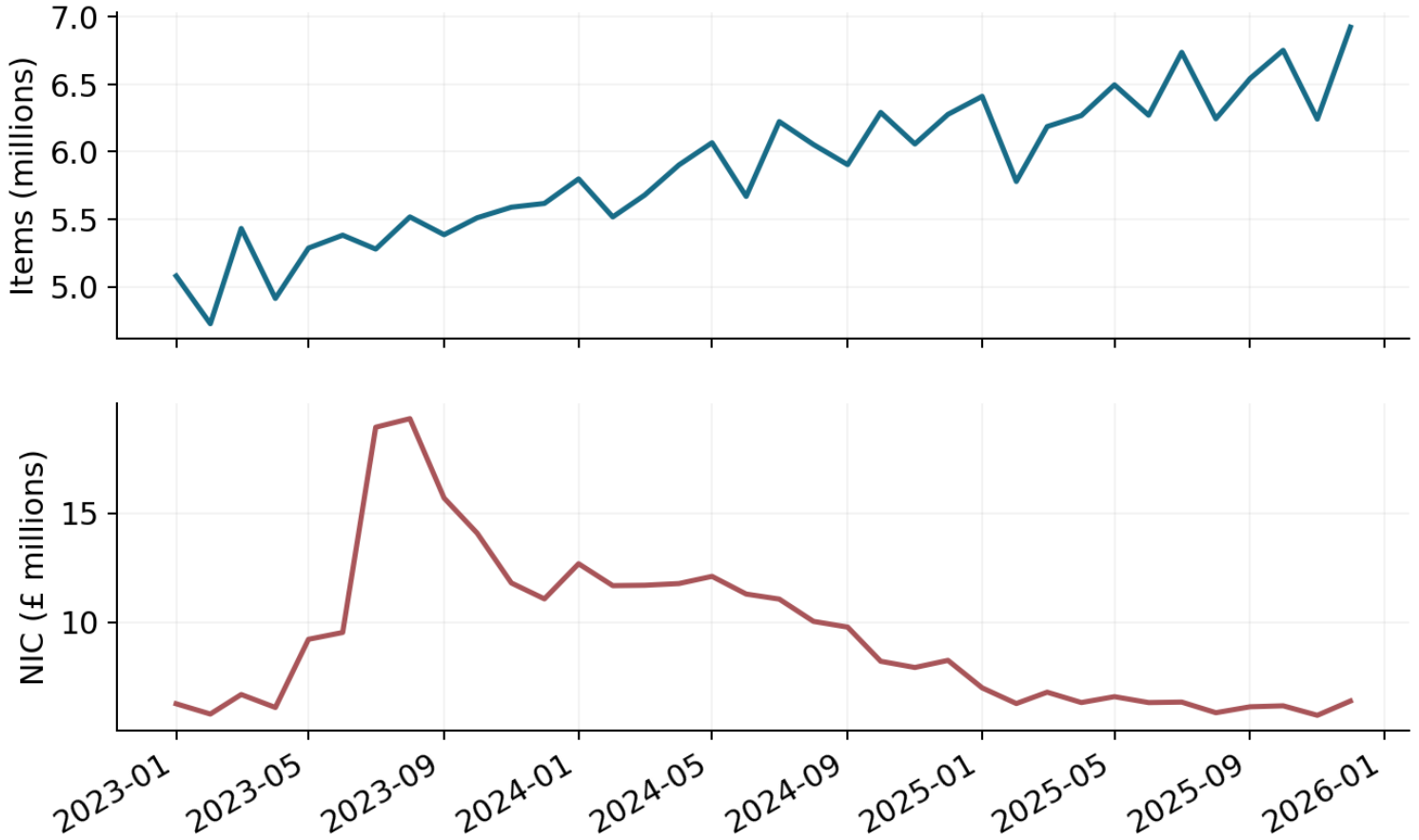
- 50,456 source rows → 3,920 monthly rows
- 7 regions · 17 presentations · 36 observed months
- Missing activity stays unknown; physical units remain separate

A traceable path from source to screen



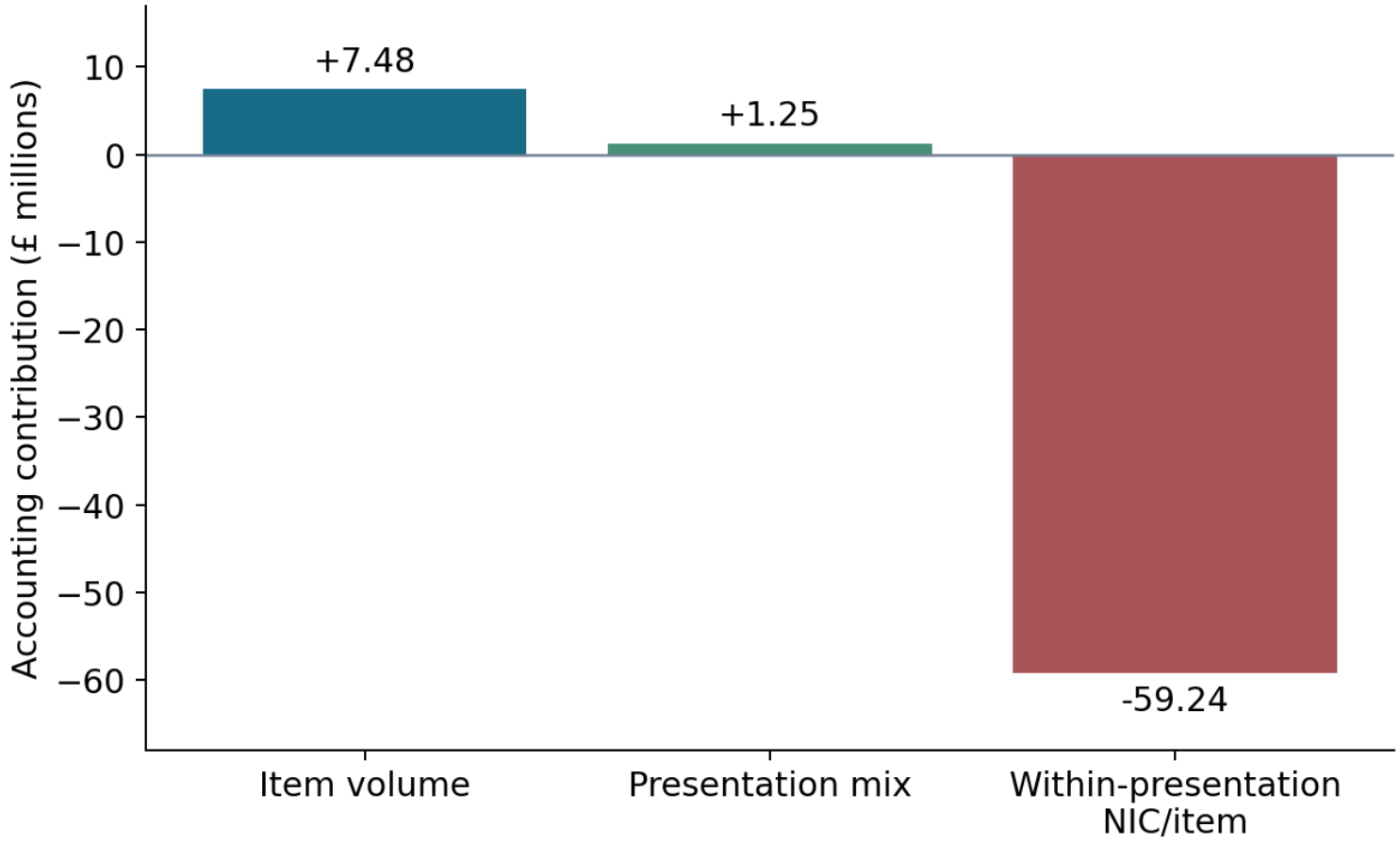
Activity rose while NIC fell

- 2025 items: 76.86m · +7.58% versus 2024
- 2025 NIC: £75.97m · −39.93% versus 2024
- The measures have different accounting meanings



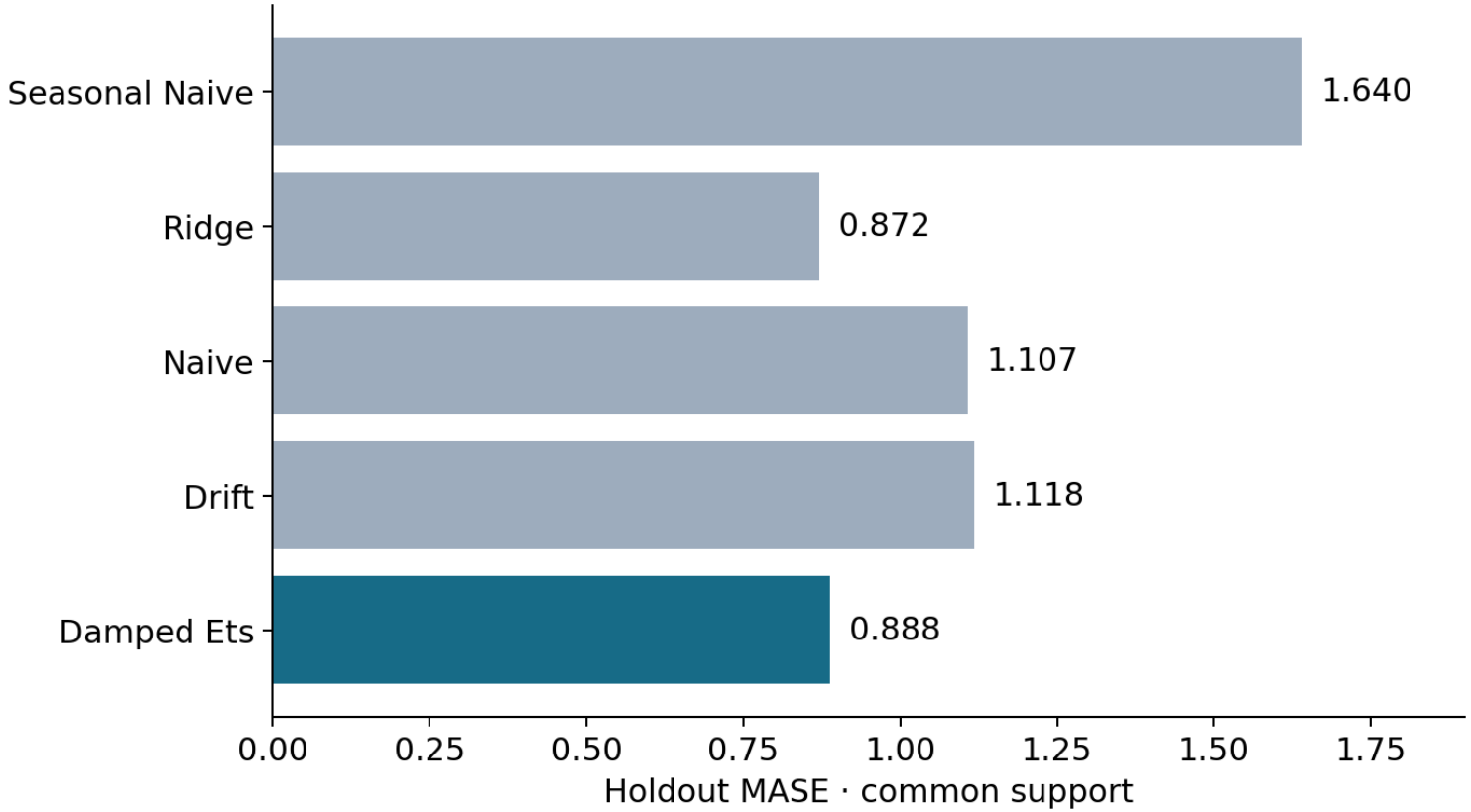
An exact identity, with a causal boundary

- Volume: +£7.48m · mix: +£1.25m
- Within-presentation NIC/item: −£59.24m
- Total observed NIC change: −£50.51m



Preserve the frozen selection

- Development selection frozen before holdout
- National ETS: MASE 0.888 · WAPE 3.684%
- Ridge comparator MASE 0.872 is reported honestly



Future-dated does not mean already validated

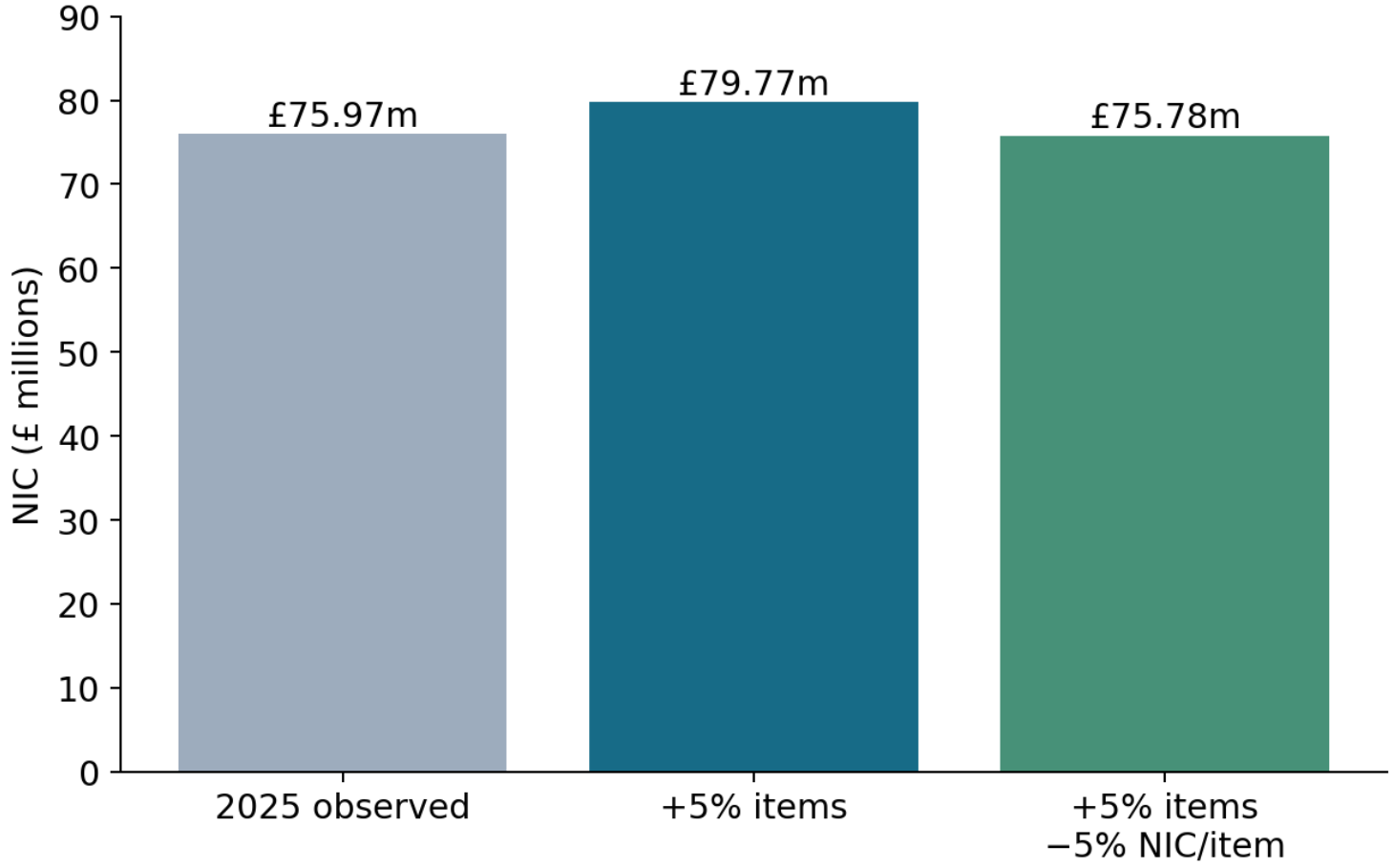
- 36 modeled series · 15 require manual review
- Intervals unavailable: residual support below stated threshold
- Jan–Mar 2026 forecasts use retrospective Dec 2025 cutoff

Rank analytical review questions

- Five weighted percentile components; policy is illustrative
- All 7 regional ranks unstable under saved sensitivity rule
- 13 presentation priorities unscored; 225 proposed investigations

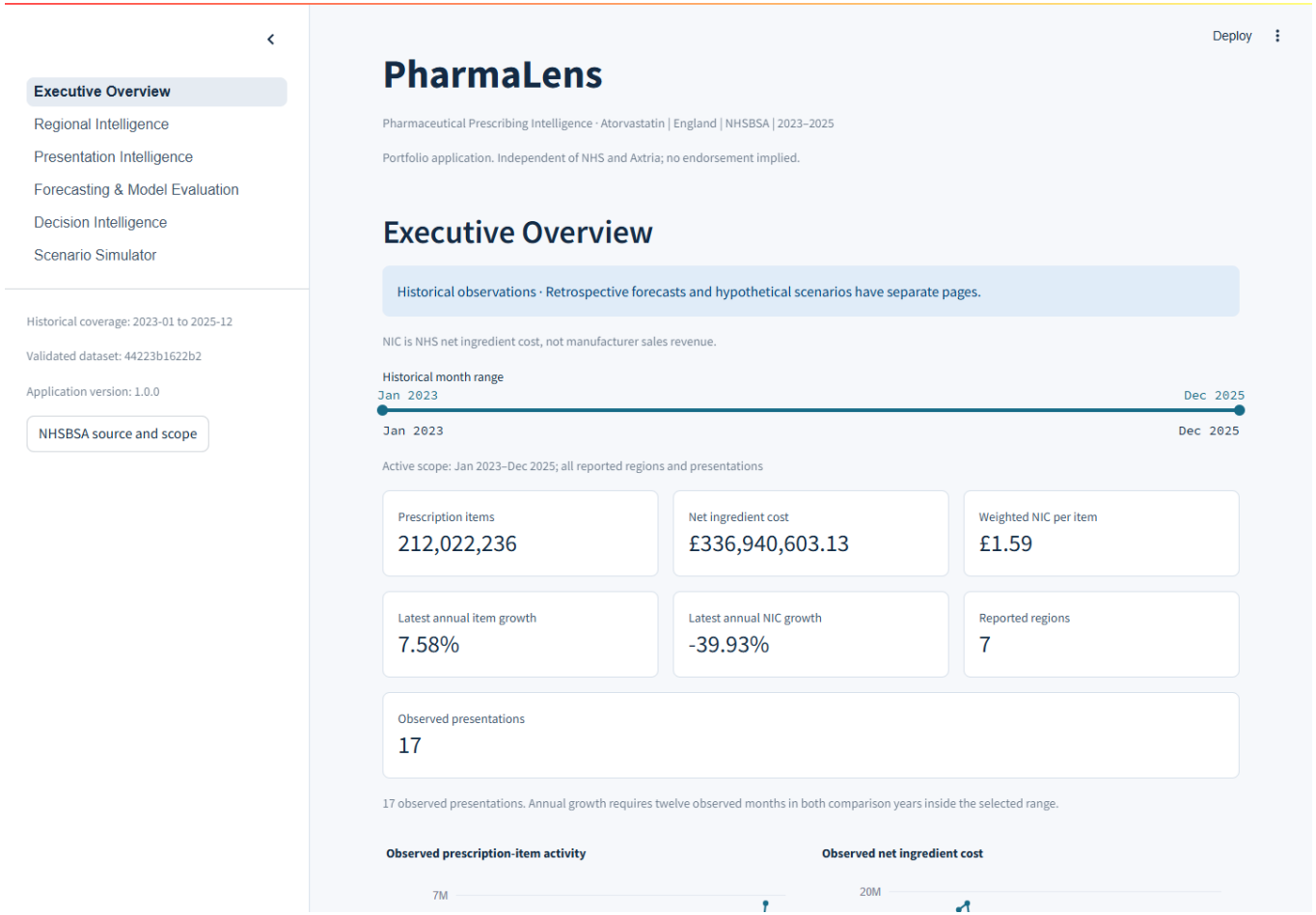
Check assumptions through accounting

- Hypothetical NIC = baseline × volume factor × ratio factor
- +5% and −5% produce −0.25%, not break-even
- Exact offset for +5% volume: −4.7619% NIC/item



Inspect the evidence, then drill down

- Six pages with scope-aware exports and reliability labels
- Independent reconciliation and protected-artifact checks
- Real browser and separate runtime-environment evidence



A defensible portfolio, with clear boundaries

- No manufacturer revenue, causal savings or clinical recommendations
- More history, release-aware evaluation and user-reviewed policies
- GitHub/cloud publication and code licence remain owner decisions